UNIT 4

ELECTROMAGNETIC INDUCTION AND ALTERNATING CURRENT

"Nature is our kindest friend and best critic in experimental science if we only allow her intimations to fall unbiased on our minds" — Michael Faraday

Learning Objectives

In this unit, the student is exposed to

- the phenomenon of electromagnetic induction
- the application of Lenz's law to find the direction of induced emf
- the concept of Eddy current and its uses
- the phenomenon of self-induction and mutual-induction
- the various methods of producing induced emfs
- the construction and working of AC generators
- the principle of transformers and its role in long distance power communication
- the notion of root mean square value of alternating current
- the idea of phasors and phase relationships in different AC circuits
- the insight about power in an AC circuit and wattless current
- the understanding of energy conservation during LC oscillations

4.1 ELECTROMAGNETIC INDUCTION

4.1.1 Introduction

In the previous chapter, we have learnt that whenever an electric current flows through a conductor, it produces a magnetic field around it. This was discovered by Christian Oersted. Later, Ampere proved that a current-carrying loop behaves like a bar magnet. These are the magnetic effects produced by the electric current.

Physicists then began to think of the converse effect. Is it possible to produce an electric current with the help of a magnetic field? A series of experiments were conducted to establish the converse effect. These experiments were done by Michael Faraday of UK and Joseph Henry of USA, almost simultaneously and independently. These attempts became successful and led to the discovery of the phenomenon, called Electromagnetic Induction. Michael Faraday is credited with the discovery of electromagnetic induction in 1831.

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